Example 1 - rivt Doc

Attn: **user example**

project: **0001**

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1i. Summary

This rivt example calculates the maximum stress and deflection in a simply supported, uniformly loaded beam. It also serves as an annotated example of a doc with multiple sections that is not part of a report.

The example illustrates the use of some of the most common API functions, commands and tags. Further details are provided in the rivt user manual.

The file may be formatted as a text, PDF or HTML doc by changing the type parameter in the PUBLISH command of the Doc API (rv.D) at the end of the file. Published files are found in the sub-folders of the _published folder.

2i. Load Combinations

Table 1: ASCE 7-05 Load Effects (saved as csv in _stored folder)

Equation No.	Load Combination
16-1	1.4(D+F)
16-2	1.2(D+F+T) + 1.6(L+H) + 0.5(Lr or S or R)
16-3	1.2(D+F+T) + 1.6(Lr or S or R) + (f1L or 0.8W)

3v. Loads and Geometry

Successive lines of value definitions are formatted as a table. Variable values are defined with the define operator. The line tag [T] labels and numbers the table.

Table 2: Define Unit Loads

variable	value	[value]	description
D_1	3.80 psf	0.18 kPA	joists DL
D_2	2.10 psf	0.10 kPA	plywood DL
D_3	10.00 psf	0.48 kPA	partitions DL
D_4	3.00 klf	43.78 kN_m	fixed machinery DL
L_1	40.00 psf	1.92 kPA	ASCE7-O5 LL
b_1	10.00 inch	254.00 mm	beam width
h_1	18.00 inch	457.20 mm	beam depth
E_1	29000.00 ksi	199947.96 MPA	modulus of elasticity

The VALTABLE command reads variable values from a file in the src folder. The description is used as the table title. The range specifies the starting and ending line to be read from the file (0:0 means all lines).

Table 3: Beam Geometry from file: **src/beam1.csv**

variable	value	[value]	description
spc_1	2.00 ft	0.61 m	beam spacing

variable	value	[value]	description
spn_1	16.00 ft	4.88 m	beam span

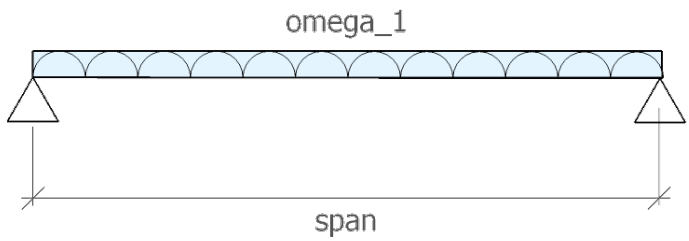

Fig. 1 - Beam Diagram

Uniform Distributed Loads

Eq. 1: dead load [ASCE7-05 2.3.2]

$dl_1 = 1.2 \cdot (D_4 + spc_1 \cdot (D_1 + D_2 + D_3))$

$dl_1 = 3.64$ klf $[dl_1] = 53.09$ kN_m | dead load [ASCE7-05 2.3.2]

D_4	spc_1	D_3	D_1	D_2
3.00 klf fixed machinery DL	2.00 ft beam spacing	10.00 psf partitions DL	3.80 psf joists DL	2.10 psf plywood DL

Eq. 2: live load [ASCE7-05 2.3.2]

$ll_1 = 1.6 \cdot L_1 \cdot spc_1$

$ll_1 = 0.13$ klf $[ll_1] = 1.87$ kN_m | live load [ASCE7-05 2.3.2]

spc_1	L_1
2.00 ft beam spacing	40.00 psf ASCE7-05 LL

Eq. 3: total load [ASCE7-05 2.3.2]

$w_1 = dl_1 + ll_1$

```
ω₁ = 3.77 klf     [ω₁] = 54.96 kN_m   | total load [ASCE7-05 2.3.2]

dl₁                          ll₁
──────────────────────────   ──────────────────────────
3.64 klf                     128.00 ft·psf
dead load [ASCE7-05 2.3.2]   live load [ASCE7-05 2.3.2]
──────────────────────────   ──────────────────────────
```

4v. Beam Response

The following lines import the beam geometry from an external file, calculate section properties from imported functions and calculate the maximum moment, bending stress and mid-span deflection.

Table 4: Beam functions from file: **src/sectprop.py**

Function	Docstring
rectsect(b, d)	section modulus of rectangle
rectinertia(b, d)	moment of inertia of rectangle
midspan_delta(ln, w, e, i)	mid-span deflection of simply supported beam with UDL

Eq. 4: rectangle - S

```
section₁ = rectsect(b₁, h₁)
```

h_1	b_1
18.00 inch	10.00 inch

section_1 = 540.00 in3	[section_1] = 8849.01 cm3	rectangle - S

Eq. 5: rectangle - I

```
inertia₁ = rectinertia(b₁, h₁)
```

h_1	b_1
18.0 inch	10.0 inch

inertia_1 = 4860.0 in4	[inertia_1] = 202288.5 cm4	rectangle - I

Eq.6

```
     M₁
σ₁ = ──
     S₁
```

Eq. 7: mid-span UDL moment

$$m_1 = \frac{\omega_1 \cdot spn_1^2}{8}$$

m_1 = 120.52 ftkip [m_1] = 163.40 mkN | mid-span UDL moment

ω_1	spn_1
3.77 klf total load [ASCE7-05 2.3.2]	16.00 ft beam span

Eq. 8: bending Stress

$$fb_1 = \frac{m_1}{section_1}$$

fb_1 = 2678.2 lb_in2 [fb_1] = 18.5 MPA | bending Stress

m_1	$section_1$
120.5 ft2·klf mid-span UDL moment	540.0 inch3 rectangle - S

Eq.9: stress ratio

$fb_1 < 20000 \cdot psqin$

fb_1	20000*psqin	ratio	check	reference
2.68 ksi	20.00 ksi	0.133908	**OK**	stress ratio

Eq. 10: mid-span deflection

δ_1 = midspan_δ(spn_1, ω_1, E_1, $inertia_1$)

omega_1	E_1	inertia_1	spn_1
3.77 klf	29000.00 ksi	4860.00 inch4	16.00 ft

delta_1 = 0.04 inch	[delta_1] = 1.00 mm	mid-span deflection

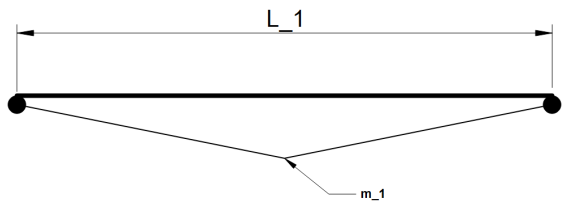

Fig. 2 - Moment Diagram

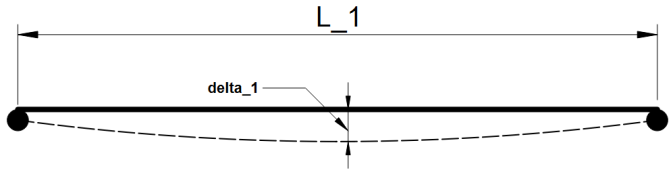

Fig. 3 - Deflection Diagram